



Final Project — Expanded Data Engineering Capstone

Path A: Official AWS Academy Capstone, expanded

Subject: Big-Data

Teacher: Agustín Villarreal Carrillo

Rol 1: Data Engineer

Jessica Abril
Quintero Castillo

Rol 2: Data Quality

Gael Loreto
Miranda

Rol 3: Analytics

Alejandro
Costich
Sandoval

Rol 4: Orchestration & Operations

César Sebastián Mercado Díaz

| Context: Sea Around Us (SAU)



The Challenge of Marine Data

We work with the global fisheries dataset spanning 1950–2018. This data is critical for understanding the human impact on marine ecosystems across international waters.

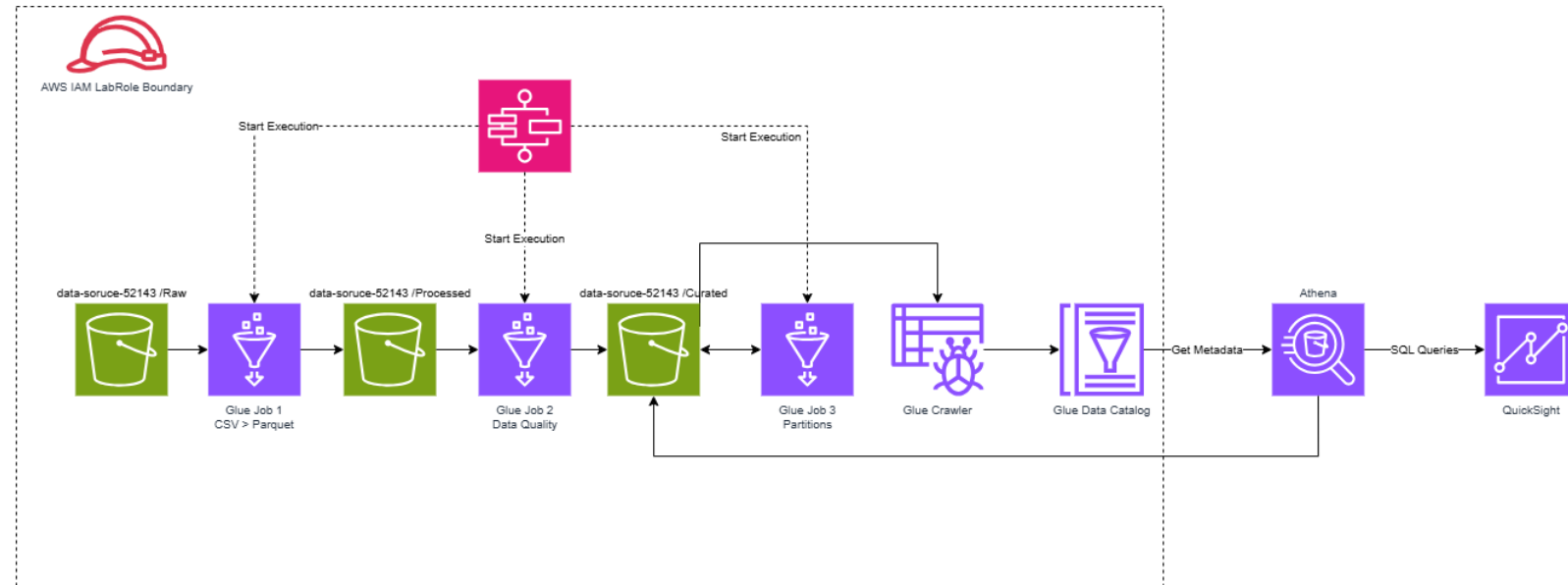
- **Sources:** Global catch, High Seas, EEZ Fiji, EEZ 848
- **Volume:** 500,000+ catch records across 68 years
- **Goal:** Transform fragmented CSV files from multiple ocean zones into a fully optimized, query-ready Data Lake on AWS

Architecture

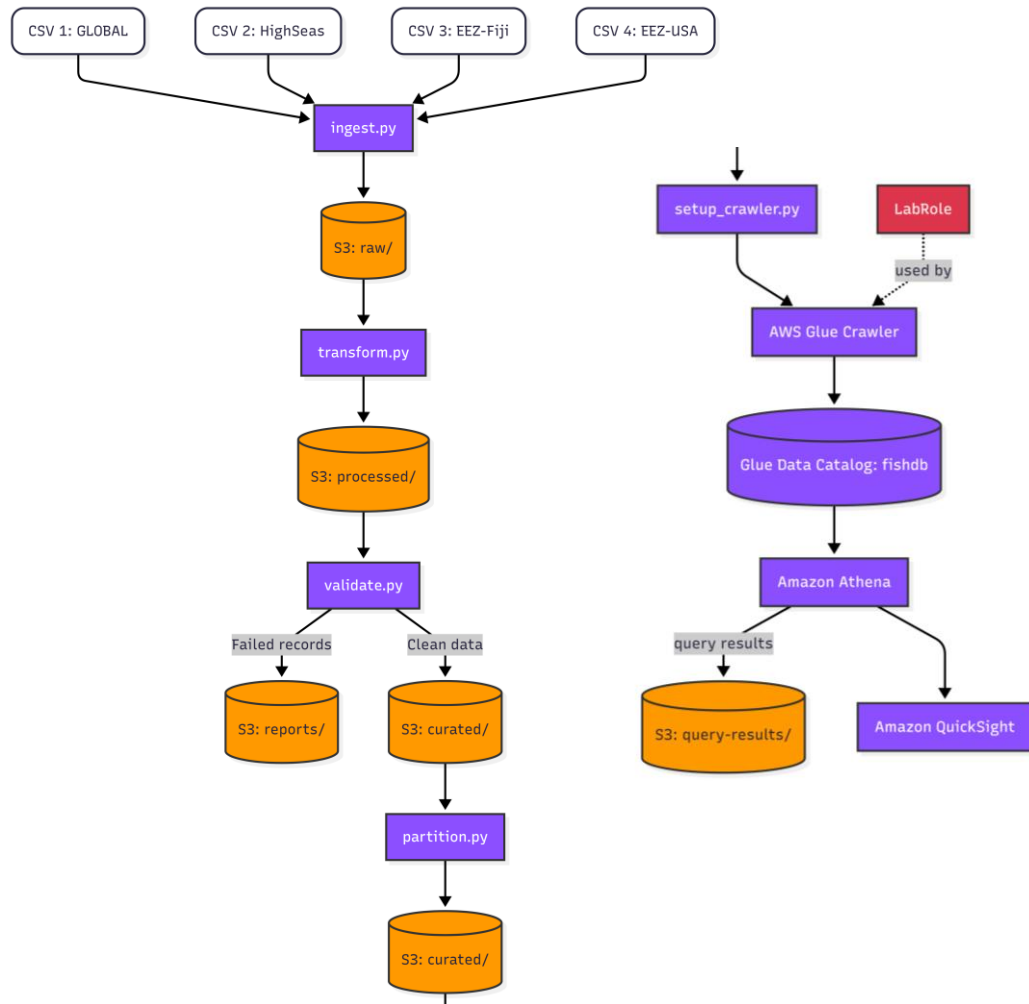
Orchestration

Our solution is 100% Serverless. **AWS Step Functions** acts as the brain of the system, triggering each Glue Job in sequence and ensuring no stage runs before the previous one completes.

- **S3 Data Lake:** Three isolated zones — data-source-52143/Raw → /Processed → /Curated
- **Glue Job 1:** Ingests raw CSVs and converts them to Parquet format
- **Glue Job 2:** Applies data quality rules before data reaches the curated zone
- **Glue Job 3:** Applies partitioning to optimize downstream query performance
- **Glue Crawler:** Scans the processed bucket and registers schema metadata into the Glue Data Catalog
- **Athena:** Queries data directly on S3 via SQL — no data movement required
- **QuickSight:** Connects to Athena to deliver the final analytics dashboard



End-to-end pipeline



Zones

- **Raw:** original csv files
- **Processed:** cleaned parquet files with aligned schemas
- **Curated:** quality-validated and partitioned by year

```
s3://data-source-52143/  
├ raw/  
├ processed/  
└ curated/
```

Scripts

ingest.py

- Creates a bucket named data-source-52143 if it does not already exist.
- Uploads the four datasets to the raw/ zone in S3.
- No data is modified at this stage.

transform.py

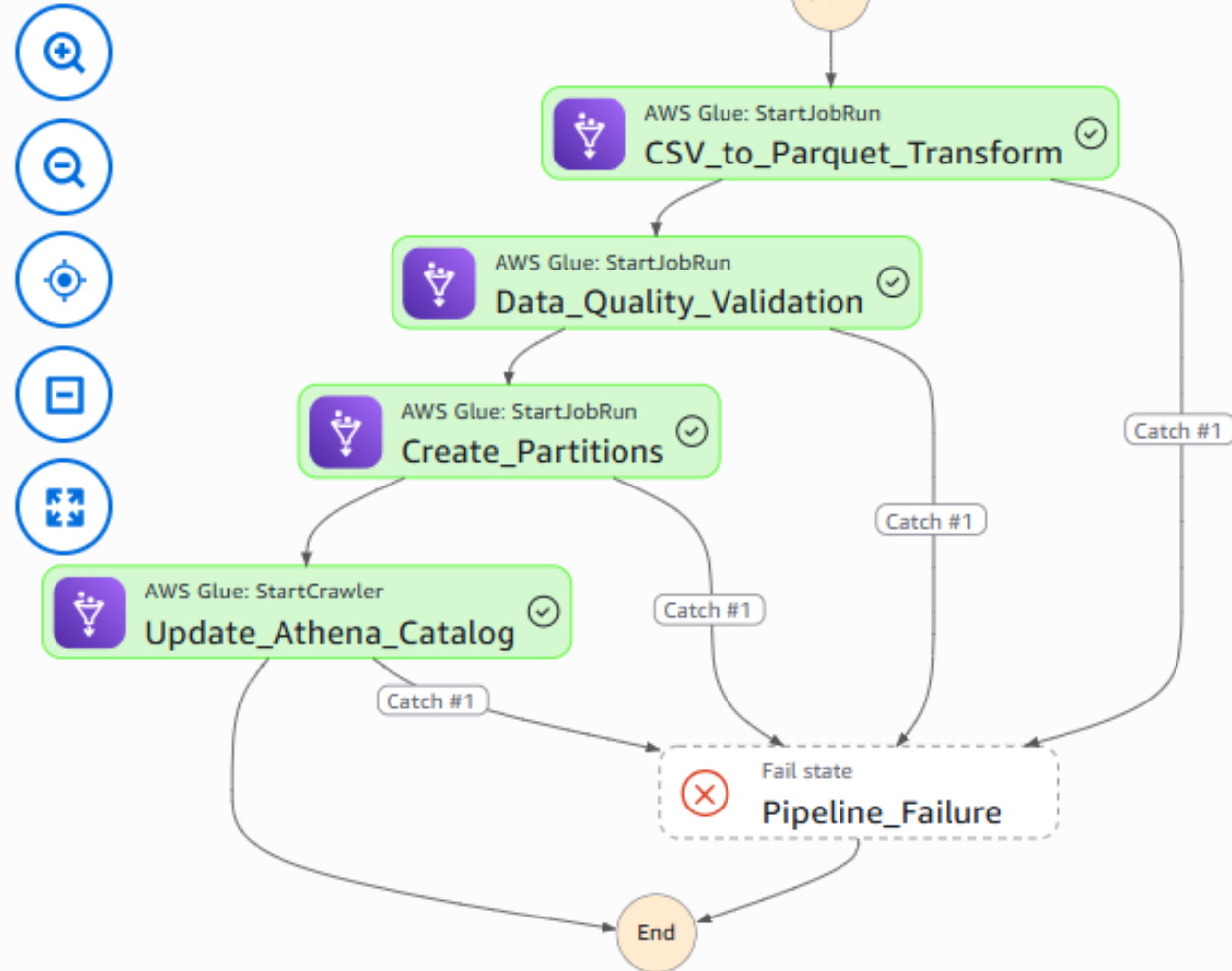
- Applies schema alignment where needed, converts to parquet format, and uploads the result to the processed/ zone.

setup_crawler.py

- Creates the fishdb Glue database and fishcrawler crawler if they do not already exist, then runs the crawler against the curated/ zone (after Role 2 validates the data and Role 3 partitions it) and waits for it to complete. Prints the list of tables created in the Glue Data Catalog.

Graph view

Actions ▼



In progress



Failed



Caught error



Canceled



Succeeded

Data Quality Report

Run: 2026-05-18 06:02:26 | Sea Around Us — Fisheries Dataset

29
Rules run

28
Passed

1
Failed

ID	Description	File	Status	Failed records	Total records	Failed %	Severity	Detail
DQ-01	Year range 1950–2018	GLOBAL	PASSED	0	561675	0.00%	Critical	
DQ-01	Year range 1950–2018	HighSeas	PASSED	0	26720	0.00%	Critical	
DQ-01	Year range 1950–2018	EEZ_Fiji	PASSED	0	27049	0.00%	Critical	
DQ-01	Year range 1950–2018	EEZ_848	FAILED	1746	83175	2.10%	Critical	[2019]
DQ-02	Tonnes >= 0	GLOBAL	PASSED	0	561675	0.00%	Critical	
DQ-02	Tonnes >= 0	HighSeas	PASSED	0	26720	0.00%	Critical	
DQ-02	Tonnes >= 0	EEZ_Fiji	PASSED	0	27049	0.00%	Critical	
DQ-02	Tonnes >= 0	EEZ_848	PASSED	0	83175	0.00%	Critical	
DQ-03	Nulls in landed_value < 20%	GLOBAL	PASSED	0	561675	0.00%	Critical	0.0% nulls
DQ-03	Nulls in landed_value < 20%	HighSeas	PASSED	0	26720	0.00%	Critical	0.0% nulls
DQ-03	Nulls in landed_value < 20%	EEZ_Fiji	PASSED	0	27049	0.00%	Critical	0.0% nulls
DQ-03	Nulls in landed_value < 20%	EEZ_848	PASSED	15289	83175	18.38%	Critical	18.4% nulls
DQ-04	Rename fish_name→common_name, country→fishing_entity	GLOBAL	PASSED	0	4	0.00%	Critical	
DQ-04	Rename fish_name→common_name, country→fishing_entity	HighSeas	PASSED	0	4	0.00%	Critical	
DQ-04	Rename fish_name→common_name, country→fishing_entity	EEZ_Fiji	PASSED	0	4	0.00%	Critical	
DQ-04	Rename fish_name→common_name, country→fishing_entity	EEZ_848	PASSED	0	4	0.00%	Critical	